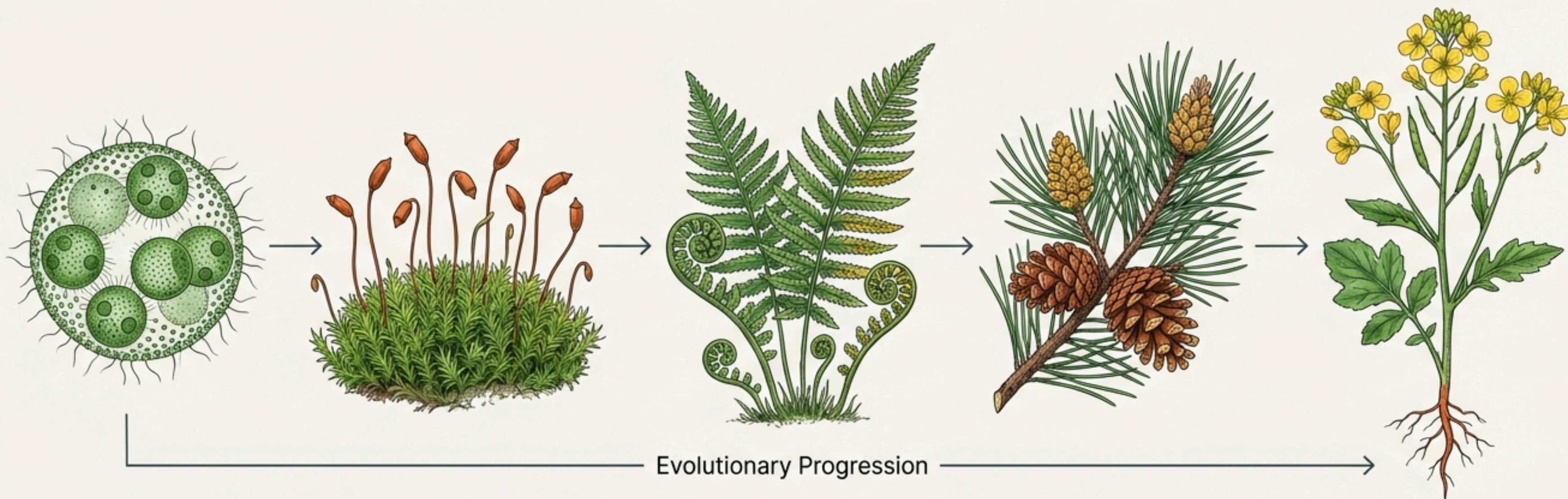


Plant Kingdom

NEET-UG Botany | Chapter 3 | NCERT-Based High Yield Review



Systems of Classification

Artificial System Linnaeus

Based on gross superficial morphological characters (habit, color, leaf number/shape).

Gave equal weightage to vegetative and sexual characteristics.

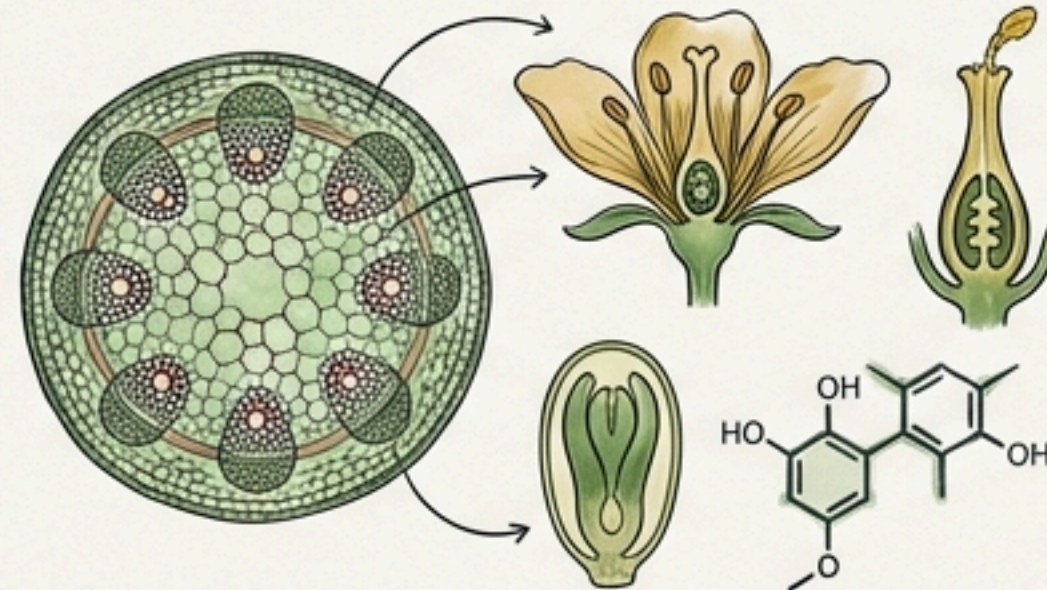


Drawback: Vegetative traits are easily affected by environment.



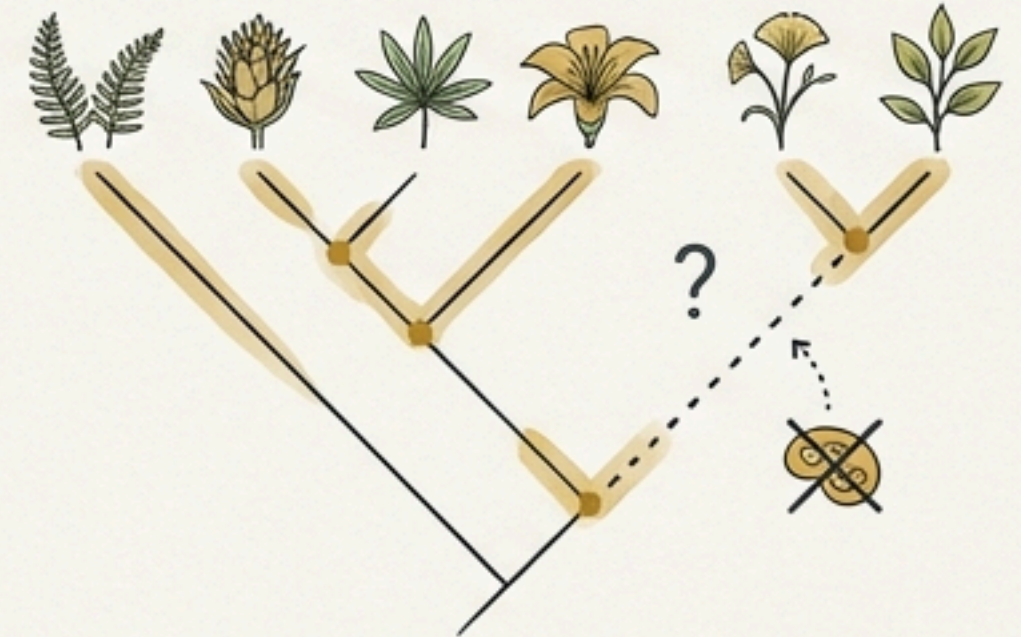
Natural System Bentham & Hooker

Based on natural affinities among organisms. Considers external features AND internal features: ultrastructure, anatomy, embryology, and phytochemistry.



Phylogenetic System Current / Hutchinson

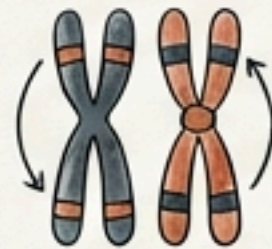
Based on evolutionary relationships. Assumes organisms in the same taxa have a common ancestor. Crucial when fossil evidence is absent.



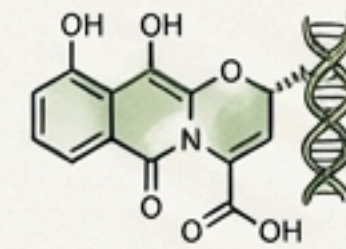
Modern Taxonomy Tools



Numerical Taxonomy: Computers assign codes to observable characters. All traits given equal importance.



Cytotaxonomy: Based on chromosome number, structure, and behavior.



Chemotaxonomy: Uses chemical constituents of the plant.

Algae: General Characteristics & Reproduction

Definition: Chlorophyll-bearing, simple, thalloid, autotrophic, largely aquatic (freshwater & marine).

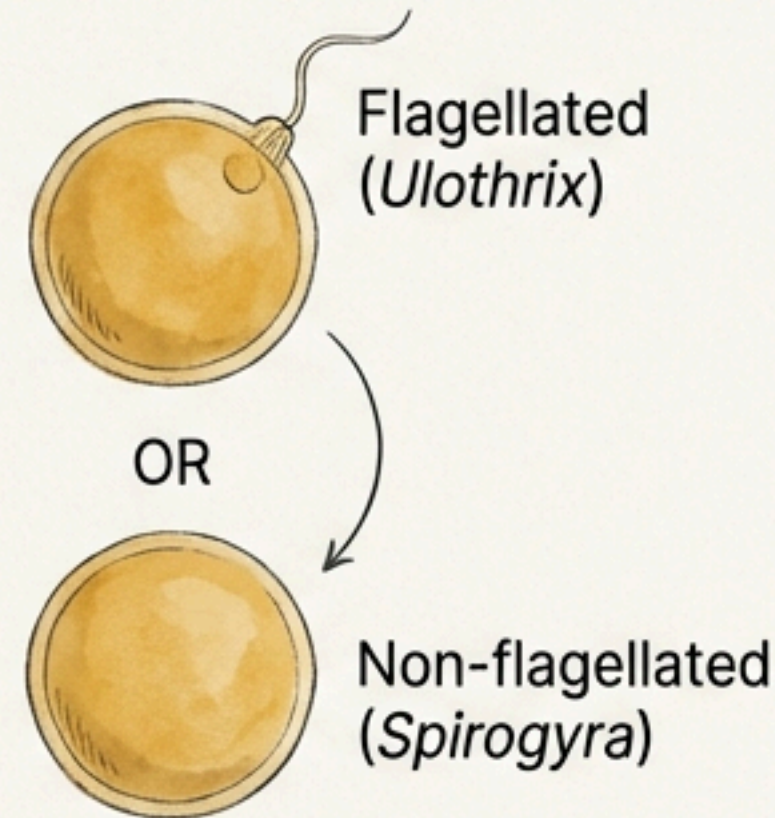
Associations: Lichen (with fungi), Sloth bear (on animals).

Economic Importance:

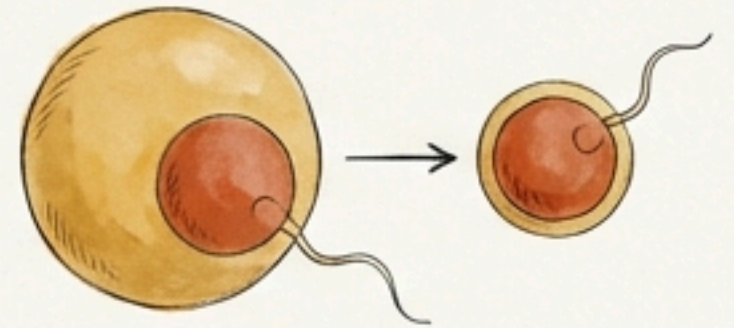
- **CO₂ Fixation:** 50% of earth's total.
- **Primary Producers:** Base of aquatic food cycles.
- **Hydrocolloids:** Algin (Brown), Carrageen (Red).
- **Commercial:** Agar (Gelidium, Gracilaria) for ice-creams/jellies.
- **Space Food:** Chlorella (protein-rich unicellular).

Sexual Reproduction Modes

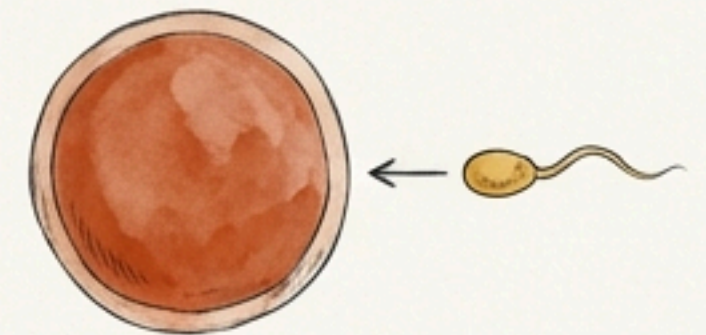
Isogamous Inter



Anisogamous Eudorina.



Oogamous Volvox, Fucus.



Vegetative Repro: Fragmentation. **Asexual Repro:** Zoospores (Flagellated).

Classification of Algae (Comparative Analysis)

NCERT Table 3.1 - High Yield for Match the Column Questions

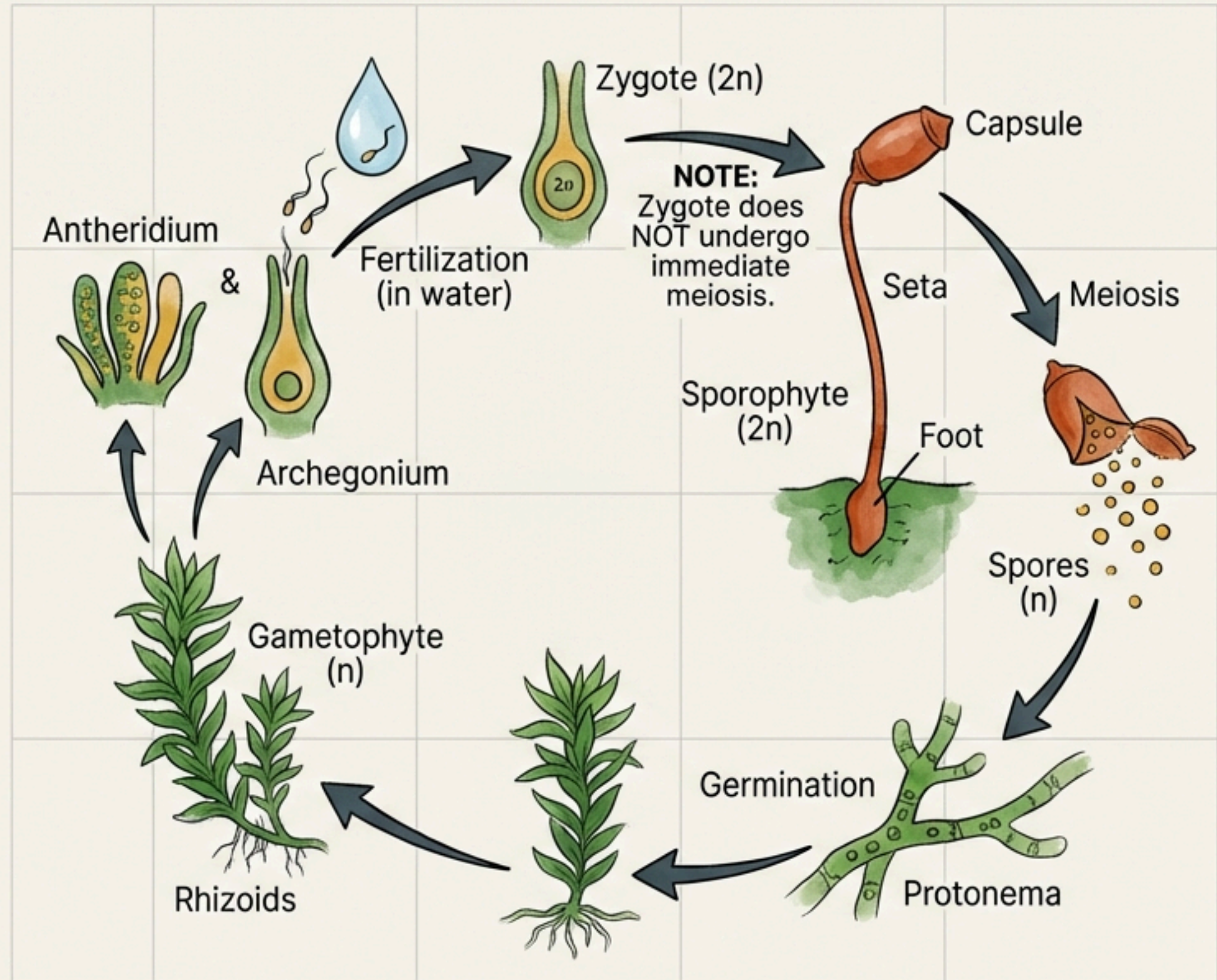
	Parameter	Chlorophyceae (Green Algae)	Phaeophyceae (Brown Algae)	Rhodophyceae (Red Algae)
1	Pigments	Chl a, b	Chl a, c, Fucoxanthin	Chl a, d, r-phycoerythrin
2	Stored Food	Starch	Mannitol, Laminarin	Floridean Starch
3	Cell Wall	Cellulose	Cellulose + Algin	Cellulose, Pectin, Polysulphate esters
4	Flagella	2-8, equal, apical	2, unequal, lateral	Absent
5	Habitat	Fresh, Brackish, Salt	Fresh (rare), Brackish, Salt	Fresh (some), Brackish, Salt (most)

Bryophytes: Amphibians of the Plant Kingdom

Defined as **Amphibians** because they can live in soil but depend on water for sexual reproduction.

Key Characteristics

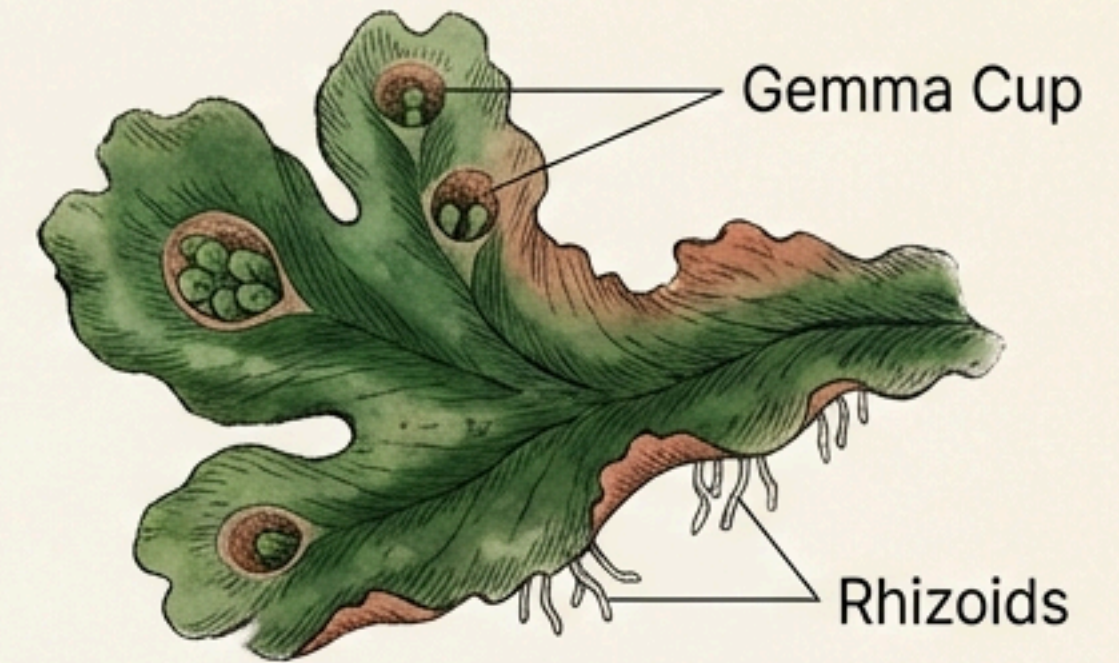
- **Dominant Phase:** Gametophyte (n). Haploid.
- **Body:** Thallus-like, attached by Rhizoids. No true root, stem, or leaves.
- **Sex Organs:** Antheridium (Male, biflagellate antherozoids) & Archegonium (Female, flask-shaped).
- **Ecological Role:** Plant succession on bare rocks. Sphagnum (Peat) used as fuel/packing material.



Diversity in Bryophytes

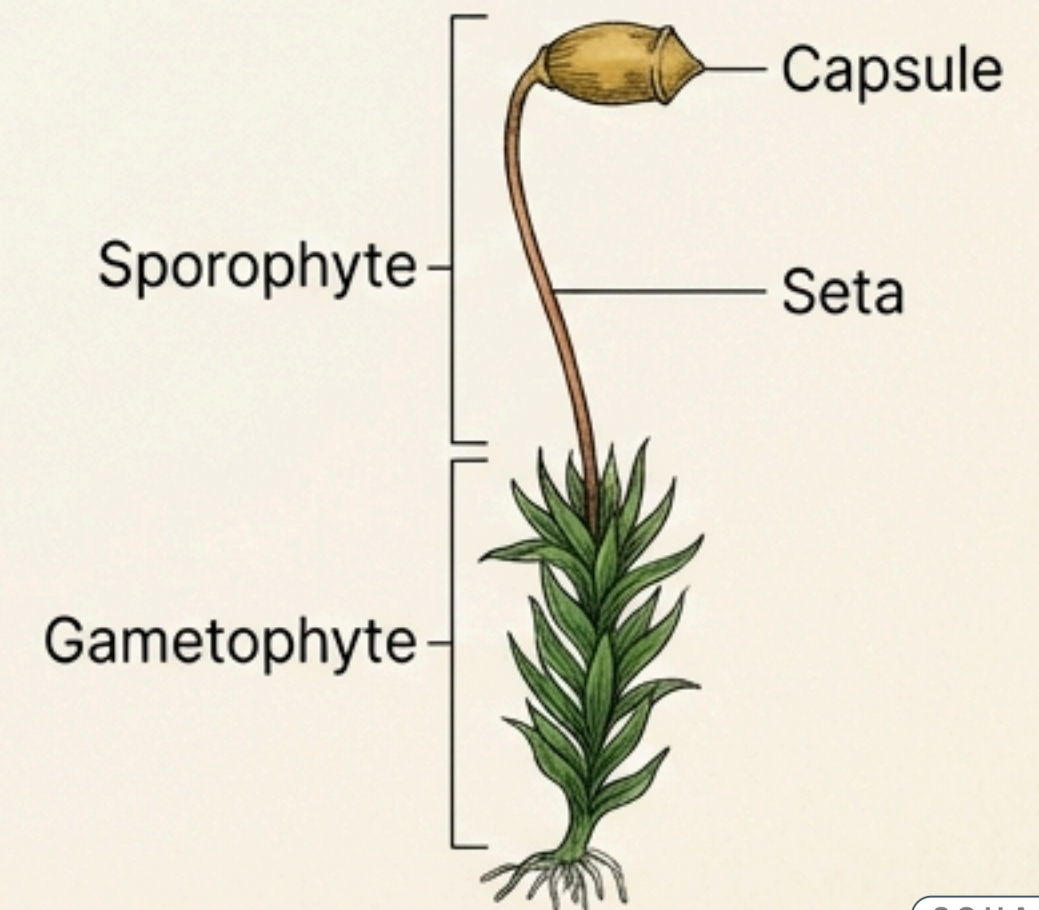
1. Liverworts (e.g., *Marchantia*)

- Thallus is dorsiventral, appressed to substrate.
- **Asexual:** Fragmentation or Gemmae (green, multicellular buds in gemma cups).
- **Sporophyte:** Foot, Seta, Capsule.



2. Mosses (e.g., *Funaria*, *Sphagnum*)

- Life Cycle has two stages: Protonema (creeping, filamentous) -> Leafy Stage (upright, spirally arranged leaves).
- **Sporophyte:** More elaborate, has spore dispersal mechanism.



Pteridophytes: The First Vascular Plants

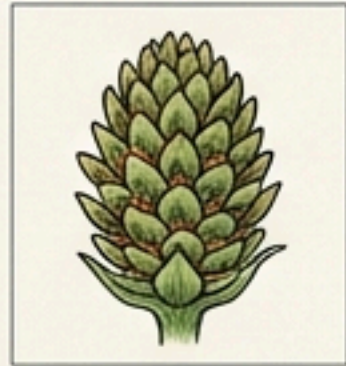
Evolutionary Milestone: Possession of Vascular Tissues (Xylem & Phloem).



Habitat

Cool, damp, shady places.
Restricted spread due to water requirement for fertilization.

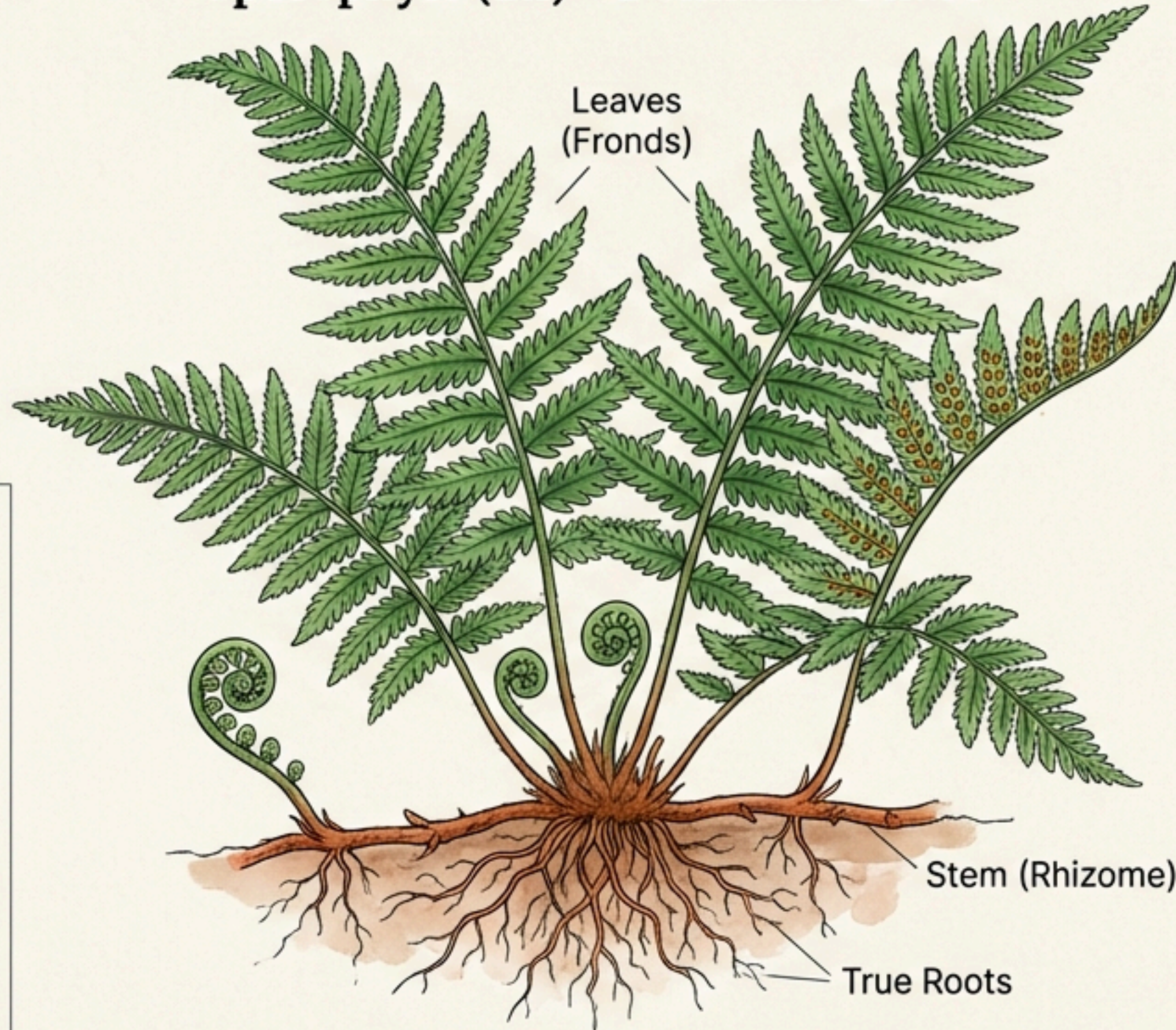
Reproduction Structures



Strobili (Cones): In *Selaginella*/*Equisetum*

Sporangia subtended by Sporophylls.
In *Selaginella*/*Equisetum*, these form compact Strobili (Cones).

Sporophyte (2n) - Dominant Phase



Sporophyte (2n) - Dominant Phase

Leaf Morphology

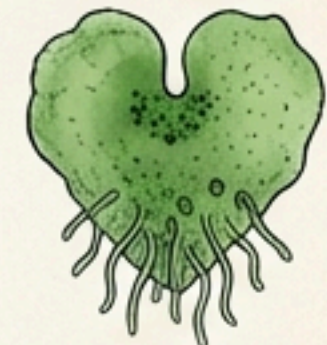


Microphylls
(Small leaves):
e.g., *Selaginella*.



Macrophylls
(Large leaves):
e.g., Ferns.

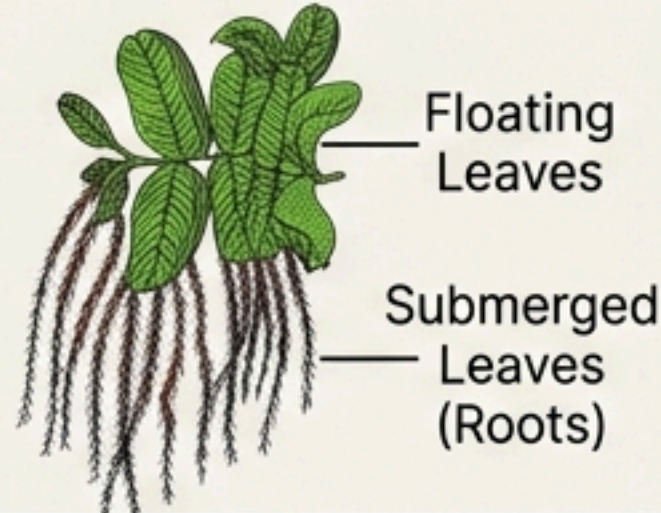
Gametophyte



Prothallus

Called Prothallus.
Small, multicellular, free-living,
photosynthetic.

Homospory vs. Heterospory & Classification

Homosporous	Heterosporous	<i>Salvinia</i>  Floating Leaves Submerged Leaves (Roots)
Spores are of similar kind. (Majority of Pteridophytes).	Spores are of two kinds – Macro (large) & Micro (small). Examples: <i>Selaginella</i> , <i>Salvinia</i> .	

Evolutionary Significance	
• Heterospory is the Precursor to Seed Habit. • Process: Female gametophyte is RETAINED on the parent sporophyte. Zygote develops into embryo WITHIN the female gametophyte.	<i>Selaginella</i> Strobilus (Cone) Macrosporangia Microsporangia Leaves (Microphylls) Stem

Classes of Pteridophytes			
Psilopsida	Lycopsida	Sphenopsida	Pteropsida
Example: <i>Psilotum</i>	Examples: <i>Selaginella</i> , <i>Lycopodium</i>	Example: <i>Equisetum</i>	Examples: <i>Dryopteris</i> , <i>Pteris</i> , <i>Adiantum</i>

Gymnosperms: 'Naked Seeds'

Ovules are not enclosed by any ovary wall; they remain exposed before and after fertilization.

Roots: Tap roots generally.

- **Pinus:** Fungal association (Mycorrhiza).
- **Cycas:** Coralloid roots (N₂-fixing cyanobacteria association).

Leaves: Xerophytic adaptations.

- Needle-like (Conifers) to reduce surface area.
- Thick cuticle & sunken stomata to reduce water loss.

Stem: Unbranched (*Cycas*) vs Branched (*Pinus*, *Cedrus*).

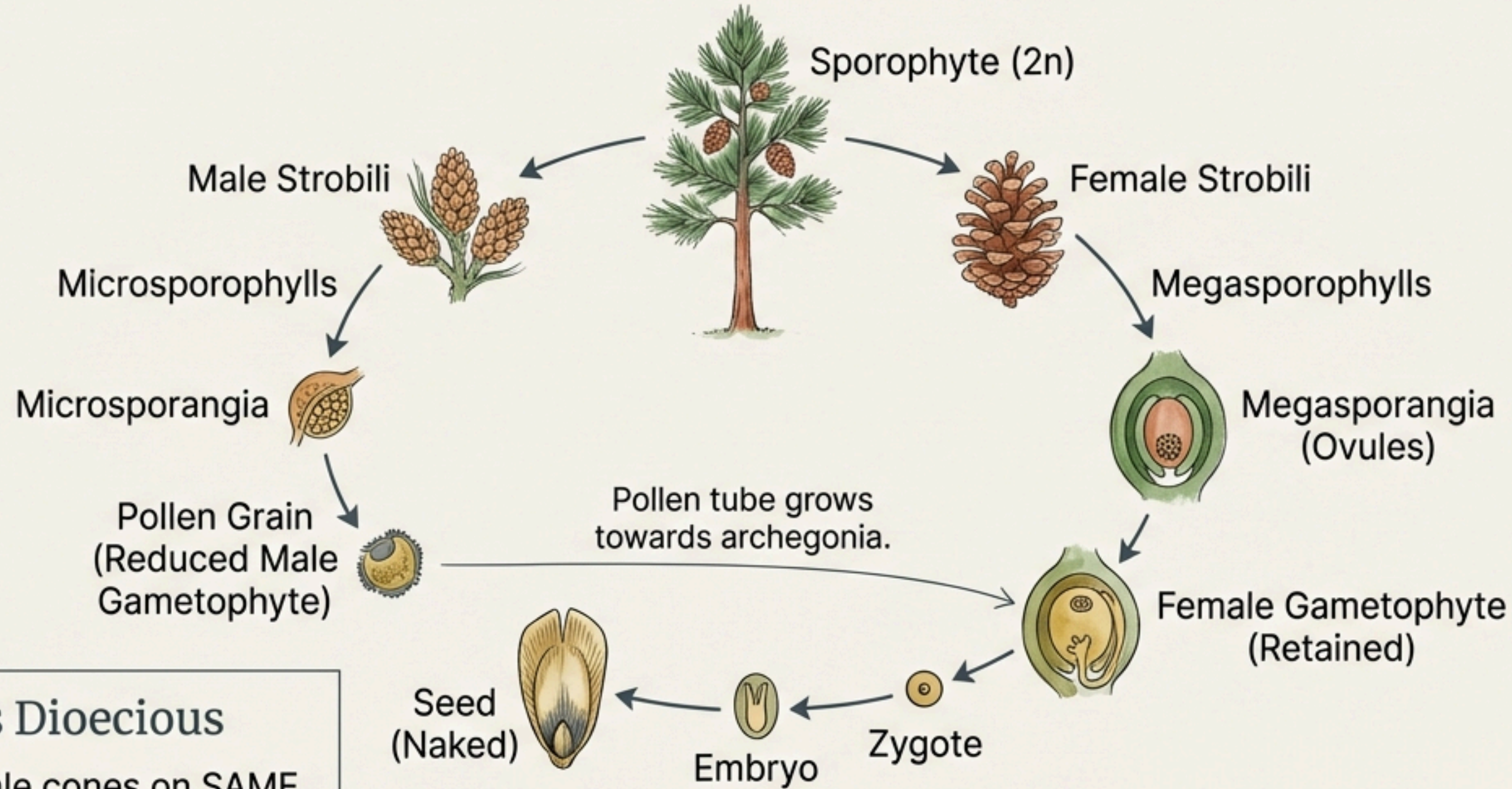
Pinus Tree



Cycas Plant

Gymnosperm Reproduction

Key Concept: Gametophytes are NOT independent.



Monoecious vs Dioecious

Pinus: Male & Female cones on SAME tree (Monoecious).

Cycas: Male cones & Megasporophylls on DIFFERENT trees (Dioecious).

Unlike Bryophytes/Pteridophytes, Gymnosperm gametophytes do not have an independent free-living existence.

Angiosperms: Flowering Plants

Key Distinction: Pollen and ovules develop in Flowers. Seeds are enclosed in Fruits.

Classification

- **Dicotyledons (Dicots):** Two cotyledons in seed.
- **Monocotyledons (Monocots):** One cotyledon in



Dicot Flower (Calendula)



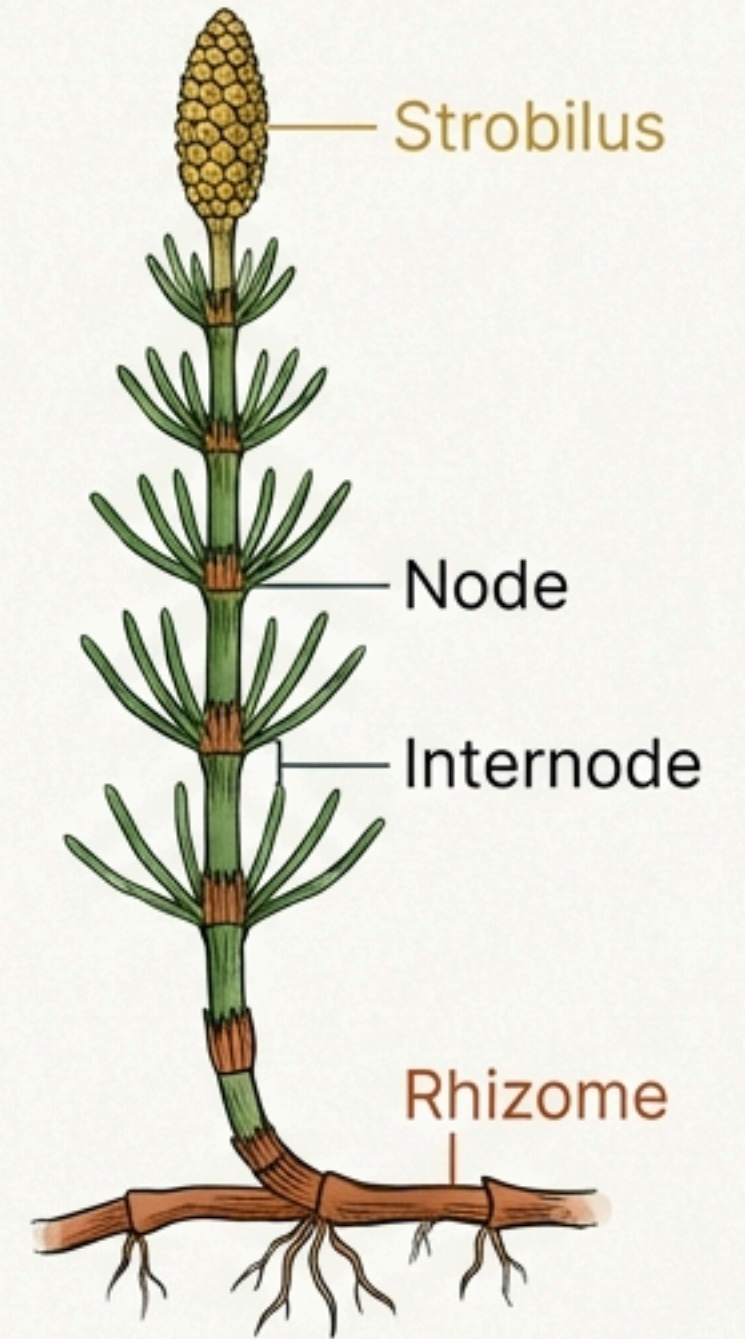
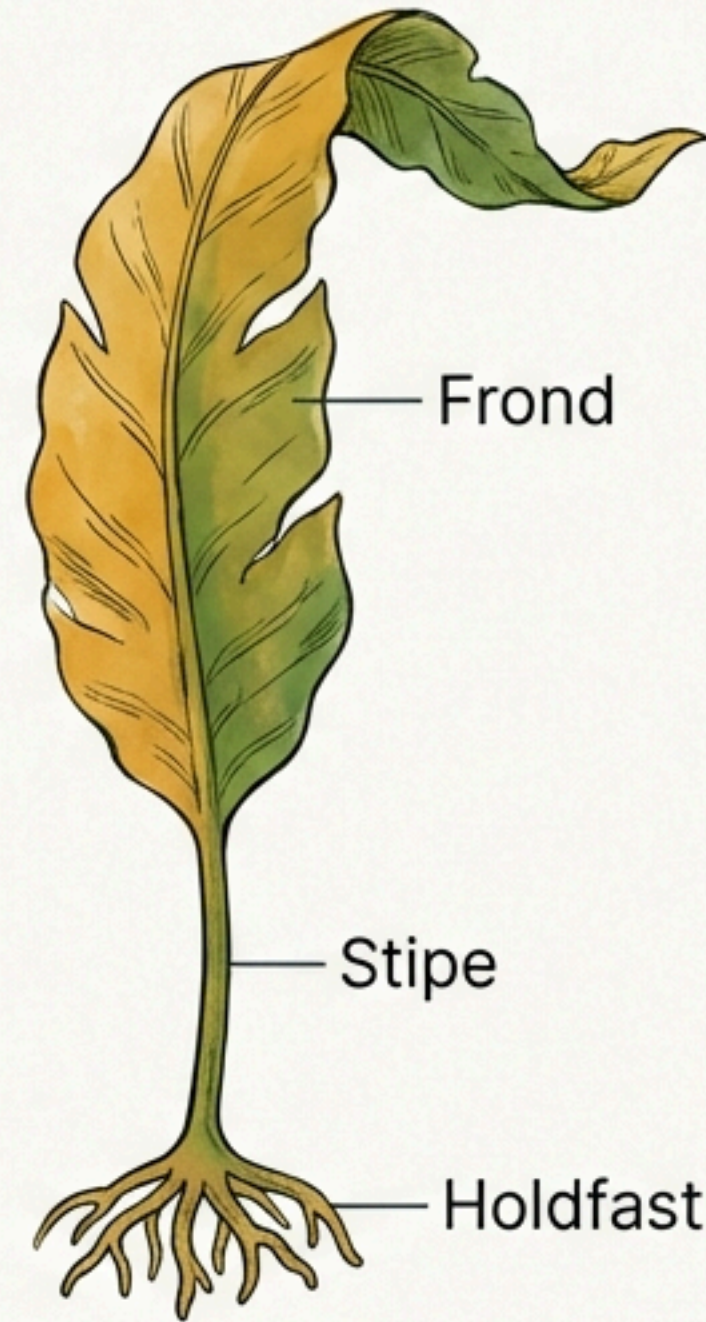
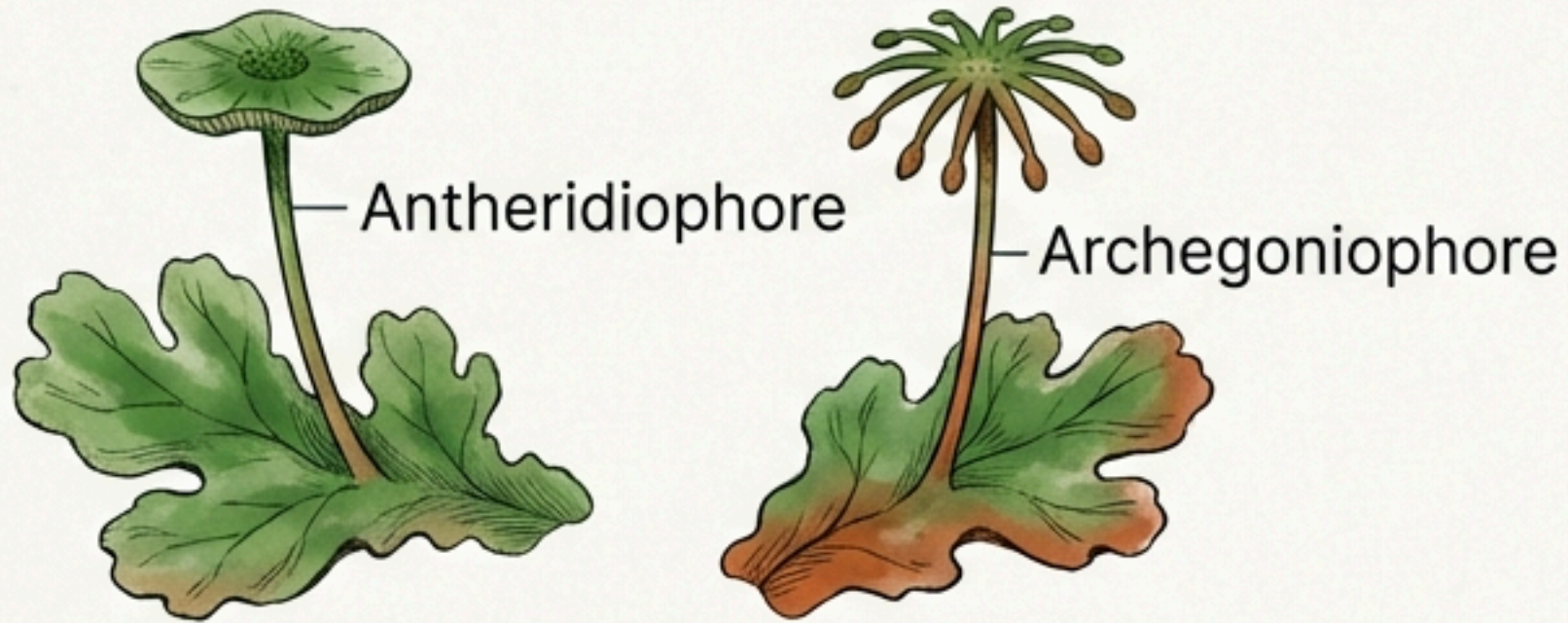
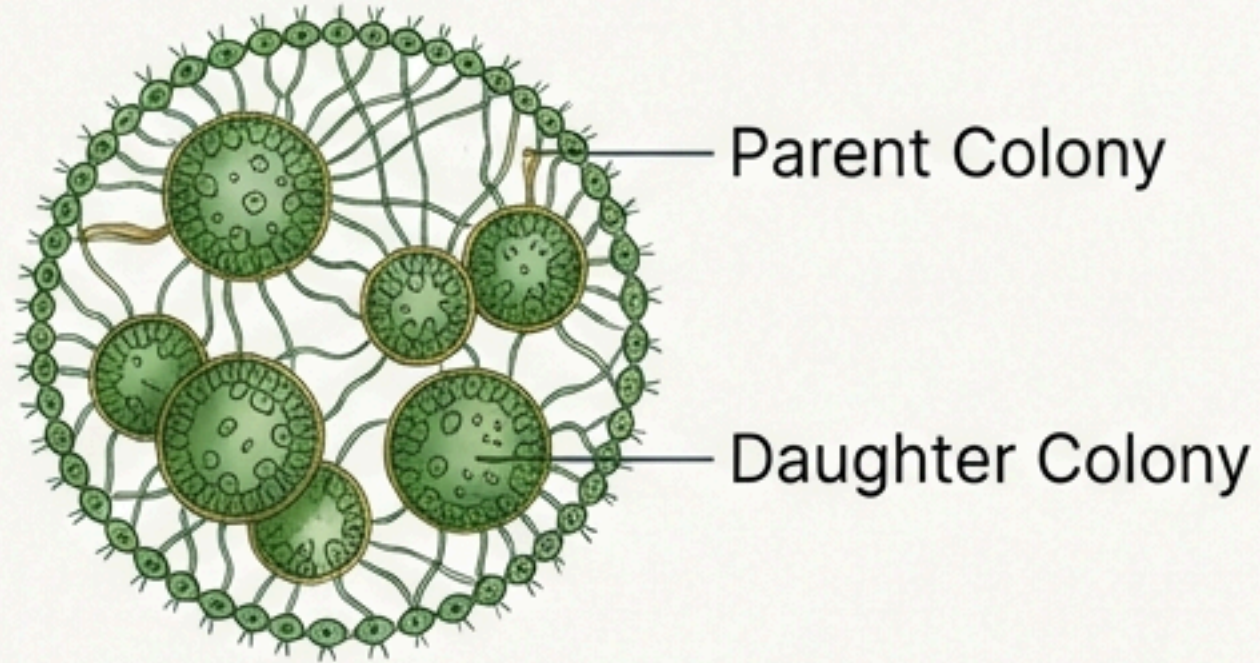
Monocot Flower (Grasses)

General Features

- **Size Range:** Tiny Wolffia to Tall Eucalyptus (>100m).
- **Reproductive Organs:** Stamen (Male) and Pistil (Female).
- **Double Fertilization:** Unique event involving syngamy and triple fusion.

Important Diagrams & Labelling (NEET Focus)

Study these specific labels carefully.



NCERT Line Emphasis

Direct Pickups for True/False Statements

- “Vegetative characters are more easily affected by environment.” (Reason for rejecting artificial classification).
- “Zygotes [in Bryophytes] do not undergo reduction division immediately.”
- “Genera like Selaginella and Salvinia... are known as heterosporous.”
- “The female gametophytes in these plants [Heterosporous Pteridophytes] are retained on the parent sporophytes for variable periods.”
- “In gymnosperms, the male and female gametophytes do not have an independent free-living existence.”

Critical Comparisons & Differences

Dominant Phase

Bryophytes:	Gametophyte (Haploid)
Pteridophytes:	Sporophyte (Diploid)
Gymnosperms:	Sporophyte (Diploid)
Angiosperms:	Sporophyte (Diploid)



Gametophyte Independence

Bryophytes:	Independent (Dominant)
Pteridophytes:	Independent (Prothallus)
Gymnosperms:	Dependent (Retained)
Angiosperms:	Dependent (Retained)



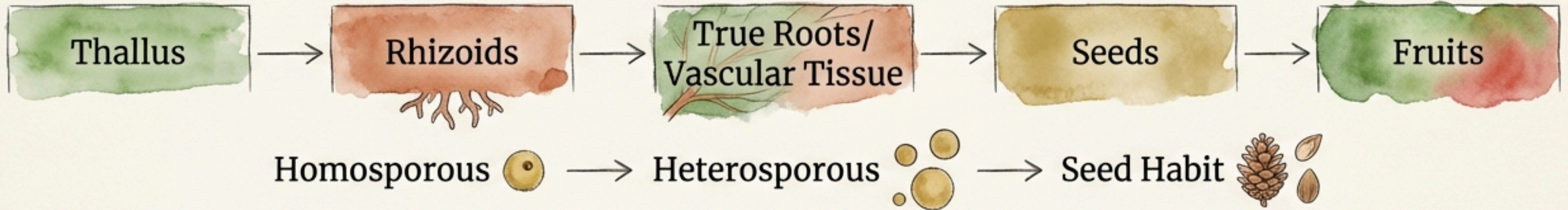
Cell Wall (Algae)

Green:	Cellulose + Pectose
Brown:	Cellulose + Algin
Red:	Cellulose + Pectin + Polysulphate esters



NEET Focus & Final Snapshot

The Evolutionary Trend (Flowchart)



Assertion-Reason Logic

A: Bryophytes are called amphibians. 
R: Require water for fertilization.

True/Explain

Water is essential for gamete motility to archegonium.

A: Red algae reach maximum depths. 
R: r-phycoerythrin absorbs blue light.

True/Explain

Blue light penetrates deepest; phycoerythrin absorbs it efficiently.

Common Traps

-  *Pinus* is Monoecious; *Cycas* is Dioecious. 
-  Red Algae have NO flagella at any stage. 
-  Brown Algae flagella: Unequal & Lateral.
-  Green Algae flagella: Equal & Apical.